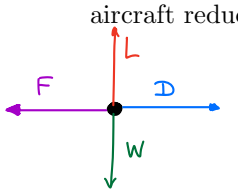
M&AE 4070/5070 - Dynamics of Flight Vehicles

Chapter 1 - Longitudinal Performances

Companion Notes for Case Study 1

This chapter reviews basics aerodynamics models and concepts, along with some major performance metrics associated with longitudinal flights: range, endurance and ceilings. Since those performances are associated with steady, horizontal flights, the governing equations associated with the movement of the aircraft reduces to a simple free body diagram:



Horizontal direction : Thrust = Drag

Vertical direction : Lift = Weight

"F" for thrust of course

$F = D$

$L = W$

1 Earth and standard atmosphere models

To simplify analysis, we always assume that the Earth:

- **is not moving**

Justification : $\dfrac{\text{Coriolis acceleration}}{\text{Acceleration of gravity}} = \dfrac{2\Omega_t V}{g} = 0.8\%$ at $M = 2$ (earth rot. rate)

- **is flat**

Justification : $\dfrac{\text{Centrifugal acceleration}}{\text{Acceleration of gravity}} = \dfrac{V^2}{R_t}\dfrac{1}{g} = 0.6\%$ at $M = 2$ (R_t : earth radius)

We also assume that **the acceleration of gravity is constant**: $g_0 = 9.806$ m/s^2.

The atmosphere is uniquely defined by its local pressure p, density ρ, and temperature T. With a few exceptions (*e.g.* constant wind speed), weather is not accounted for, and the atmosphere is assumed to be steady: p, ρ, and T are only function of the altitude h. Moreover, we will always assume that the atmosphere is **standard**, that is, is defined according to the following points:

- **the atmosphere is an ideal gas**

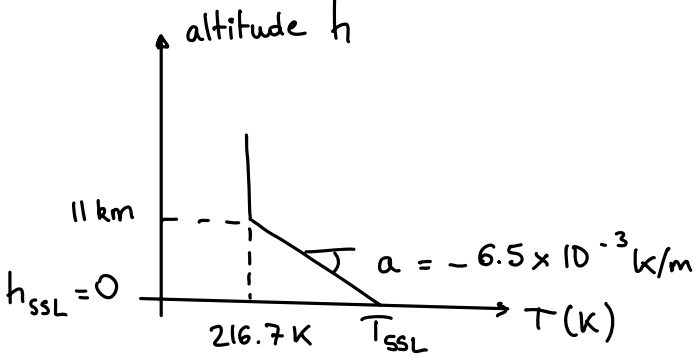
$$\frac{p}{\rho} = RT$$

(p : pressure in Pa; ρ : density in kg/m^3; T : temperature, K; R : specific ideal gas constant = 287 J/kg.K)

- **the temperature profile as function of altitude is imposed**

  - linearly decreases until 11km
  - constant above

* typical commercial aircraft altitude : ~30,000 ft (10 - 11 km)
* SR71 : 24 km



- **the atmosphere obeys the hydrostatic equation**

$$dp = -\rho g_0 dh \tag{1}$$

which, combined with a constant acceleration of gravity and the ideal gas law, results in the following profiles as function of the altitude:

Below 11 km

References are values at SSL

$$\frac{p}{p_{\text{ref}}} = \left(1 + \frac{ah}{T_{\text{ref}}}\right)^{-\frac{g_0}{aR}}$$

$$\frac{\rho}{\rho_{\text{ref}}} = \left(1 + \frac{ah}{T_{\text{ref}}}\right)^{-\frac{g_0}{aR}-1}$$

$$T = T_{\text{ref}} + ah$$

ref = SSL

Above 11 km

References are values at 11 km

$$\frac{p}{p_{\text{ref}}} = e^{-\frac{g_0}{RT_{\text{ref}}}(h-h_{\text{ref}})}$$

$$\frac{\rho}{\rho_{\text{ref}}} = e^{-\frac{g_0}{RT_{\text{ref}}}(h-h_{\text{ref}})}$$

$$T = T_{\text{ref}}$$

ref = value at 11 km

- The values at sea level (Standard Sea Level, or SSL) are:
  - $T_{SSL} = 288.15$ K
  - $P_{SSL} = 101,325$ Pa $= 1$ atm
  - $\rho_{SSL} = 1.225$ kg/m^3

Orders of magnitude

DENSITY

ρ is divided by 2 @ 6.5 km
by 3 @ 10 km

After 11 km, ρ is divided by 2 every 4.5 km.

PRESSURE

p is divided by 2 @ 6 km
by 3 @ 8.5 km
by 4 @ 10 km

After 11 km, p is divided by 2 every 4.5 km.

2 Aircraft parameters and useful models

2.1 Simplifying assumptions

In MAE 4070/5070, we only consider fixed-wing planes, and make the following simplifying assumptions:

- **Rigid bodies** Obviously questionnable !



787 Dreamliner ultimate-load wing-up bending test.

Rigid body assumption ⟺ Flexible body reality

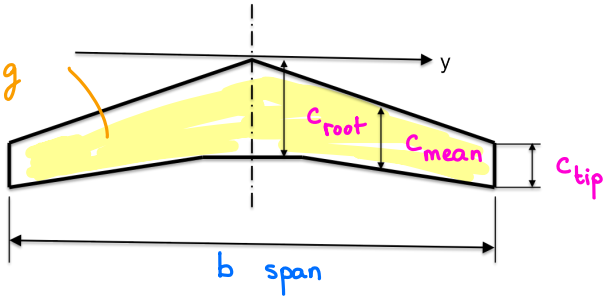
OK for small planes enough for basic flight dynamics

fluid structure interactions aero-elastic instabilities, flutter needed by structure team

- **Symmetry** with respect to vertical plane :
  - L_D includes mass*, geometry, engines**
    - * small asymmetry possible due to payload/passengers
    - ** engines rotate in the same direction (not symmetric) but effect is limited

2.2 Wings

planform area of wing S



c_{mean} or ℓ = mean aerodynamic chord

Aspect ratio : $AR = \frac{b^2}{S}$

- long, slim wings : AR is large (7-10)
- delta wings : AR is small (2-5)

2.3 Masses

We'll use the following definitions and notations:

- m_{empty}: mass of aircraft without fuel nor payload.
- m_{fuel}: typically refers to the **initial** mass of fuel.
- m_{payload}: anything on board the aircraft that is necessary to accomplish a given mission.
- m_{engine}: mass of... engines!
- $m_{\text{structure}}$: m_{empty} - m_{engine}.
- m_i: initial mass of the aircraft, including fuel (also take-off mass).
- m_f: mass of the aircraft at the end of the mission.

$$\begin{cases} m_{fuel} = m_i - m_f \\ m_f = m_{empty} + m_{payload} \end{cases}$$

It is convenient to define the following *relative* mass coefficients, which are just ratios between certain masses:

$$m_{payload} = u_m m_i \quad m_{empty} = v_m m_i \quad m_{fuel} = w_m m_i \quad \text{with} \quad u_m + v_m + w_m = 1 \quad \text{and} \quad u_m < v_m, w_m$$

Rule of thumb: v_m is around $1/2$ for most aircrafts

w_m corresponds to the percentage of its total mass in fuel an aircraft is able to carry, evaluated at take-off. It plays a very important role in determining range and endurance performance metrics.

$$m_f = (v_m + u_m) m_i = (1 - w_m) m_i$$

$w_m \approx$ 0.2 for short-haul
0.3 for long-haul
up to 0.5 for specialized surveillance aircrafts

2.4 Fuel consumption

How fuel consumption is characterized for a given engine depends on the type of engine: propeller/reciprocating engine or jet.

- For a propeller/reciprocating engine, the fuel consumption is proportional to the power delivered to the shaft: P_E

$C_P = C_{s_P} P_E$ — C_P: actual consumption, kg/h; C_{s_P}: specific consumption, kg/kW.h; P_E: delivered by the engine, kW

with $P < P_E$ (power delivered by engines) and $P = \eta_p P_E$ (η_p: efficiency coefficient)

power generated by propellers

the power generated by the propellers is the one that would need to be equal and opposite the drag x velocity for horizontal, steady flight

- For a jet engine, the fuel consumption is directly proportional to the available thrust force, **F**:

$C_J = C_{s_J} F$ — C_J: kg/h; F: thrust generated by the engines; C_{s_J}: specific fuel consumption, usually kg/daN.h

Order of magnitude

$C_{s_P} \approx 0.2$ to 0.4 kg/kW.h
$C_{s_J} \approx 0.45$ to 0.95 kg/daN.h

3 Aerodynamic forces and their coefficients

Any problem involves a number of physical parameters that are related to each other through governing equations or state relations (*e.g.* ideal gas law). These relations *have to be dimensionally homogeneous*, which imposes some additional constraints on the way the different physical parameters can be combined **regardless of the exact details of the governing equations.**

This leads to the existence of **non-dimensional groups**, *i.e.* groups of physical variables combined in such a way that units cancel each other out. These groups are very important, since **it is the value of those non-dimensional groups, rather than those of the individual physical parameters, that truly control the behavior of the problem.** Those non-dimensional groups are not unique, but some formulations are more useful than others.

The non-dimensional groups related to the aerodynamic forces and moments are called *coefficients*. In the context of this class, we consider the force/moment coefficients **in the incompressible limit**, that is, we dismiss the speed of sound in the list of relevant physical parameters. This assumption can be lifted by adding correcting factors that involve the Mach number.

General expressions for force and moment coefficients. Using Buckingham-Π theorem, we obtain the following:

aero forces (Lift L, Drag $\mathcal{D}$) $= f(V, \mathcal{L}, \rho)$ — $\mathcal{L}$: Characteristic length

Buckingham $\Pi \Rightarrow$ $L, \mathcal{D} \propto \rho V^2 \mathcal{L}^2$ — "there exists a proportionality constant"

Lift force $L = \frac{1}{2} \rho S V^2 C_z$

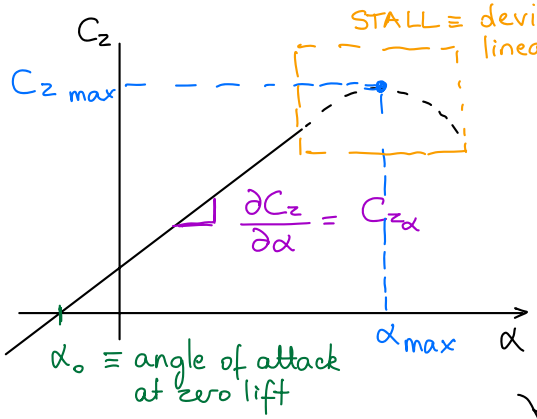
Drag force $\mathcal{D} = \frac{1}{2} \rho S V^2 C_x$

$\frac{1}{2}$ added for convenience; C: force coefficient

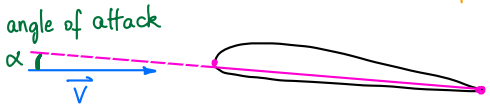
$S \equiv$ Reference surface area.
→ wing/empennage planform area when looking specifically at those lifting surfaces
→ main wing area when looking at full aircraft

3.1 A closer look at the lift coefficient

Key results from thin airfoil and Prandtl lifting line theory → lifting surface like a wing

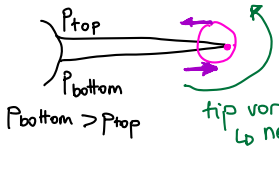


STALL ≡ deviation from linear behavior → due to flow separation on airfoil.



$C_{z_\alpha} = 2\pi$ for 2D configuration

$C_{z_\alpha} = \frac{2\pi}{1 + \frac{2}{AR}}$ for finite wing span

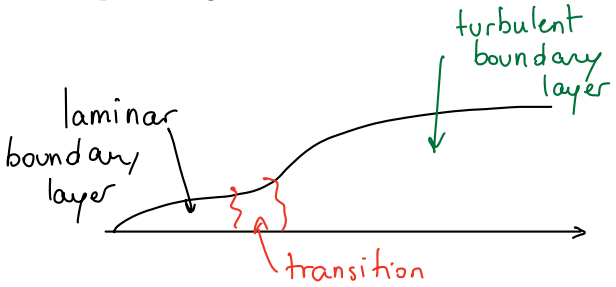


$C_{z_\alpha}\big|_{finite\ wing} < C_{z_\alpha}\big|_{2D,\ infinitely\ long\ wing}$

3.2 A closer look at the drag coefficient

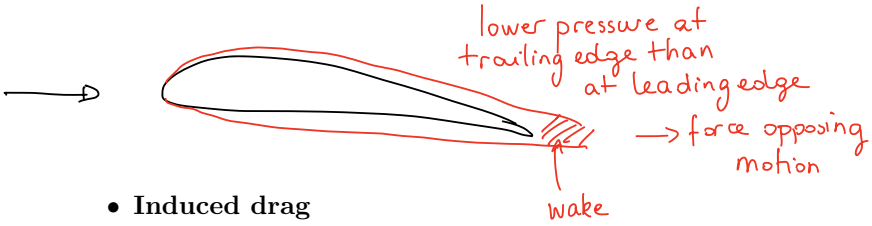
The sources of drag are:

- **Viscous effects on the aircraft surface, or skin friction drag**. The boundary layer theory provides good estimates for it.



$$C_{x,\text{viscous}} \simeq \text{constant} = a$$

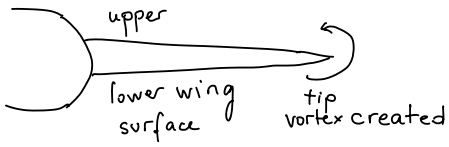
- **Pressure/form drag**, coming from the wake. This contribution is difficult to estimate, and is modeled as an affine function of the lift coefficient:



$$C_{x,\text{pressure}} = b + c\,C_z^2$$

constant part; lift contribution

- **Induced drag**



$$C_{x,\text{induced}} = d\,C_z^2$$

Total (incompressible) drag and the drag polar

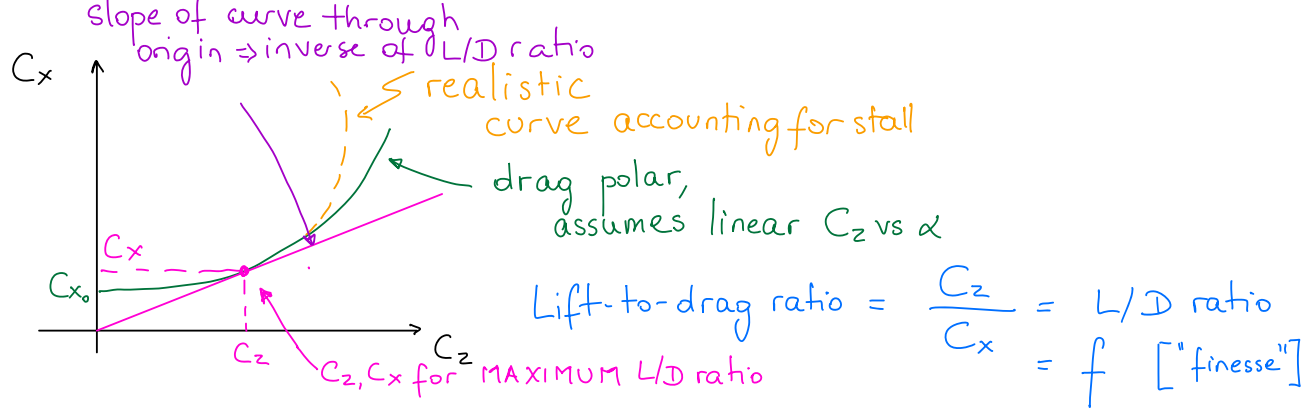
ρ = constant

$$C_x = a + b + (c+d)C_z^2 \qquad (a < 0)$$

$$C_x = C_{x_0} + k\,C_z^2$$

$$k \approx \frac{1}{\pi\, e\, AR}$$

e empirical; often $= 1$ in this class



$$\text{Lift-to-drag ratio} = \frac{C_z}{C_x} = \text{L/D ratio} = f\ [\text{"finesse"}]$$

4 Cruise performance metrics: maximum range and endurance

The two main cruise performances are 1) the capability to go far (Range, R), and 2) the capability to fly for a long time (Endurance, E). Endurance and Range can be easily estimated using so-called Breguet formulas, which are constructed using simple physical arguments. Those formulas depend on the type of engine, and to a smaller extend, how the pilot will choose to fly (*control laws*).

- Endurance: flight time ⇒ depends on how much fuel you have and how fast you consume it

fuel consumption $C = -\frac{dm}{dt} \rightarrow dt = -\frac{dm}{C}$ (mass change for aircraft ONLY comes from fuel consumption)

$$E = \int_{t_i=0}^{t_f} dt = -\int_{m_i}^{m_f} \frac{dm}{C}$$

(E: time; m_i: see above)

- Range: distance flown

Proper quantity to consider is $\frac{dm}{dx} = \frac{dm}{dt}\frac{dt}{dx} = \frac{1}{V}\frac{dm}{dt} = -\frac{C}{V}$ (mass of fuel consumed per unit distance flown)

$$R = \int_{t_i=0}^{t_f} V\,dt = -\int_{m_i}^{m_f} \frac{V}{C}\,dm$$

(R: distance)

Range and endurance can be calculated for any aircraft configuration. But when calculating performances, we are actually interested in the **maximum** values of range and/or endurance a given aircraft can achieve with a given amount of fuel. Therefore, as we integrate the inverse of the fuel consumption (time) rate (to calculate endurance), or the fuel consumption (distance) rate (to calculate range), we will also figure out the various cruise parameters that will maximize how much time the aircraft is able to stay in the air, or how much distance the aircraft can fly. The first objective is to identify what those cruise parameters are.
Note: The mass of the aircraft can obviously NOT be assumed constant when evaluating range and endurance. This contrasts with the rest of the class, where mass will be treated as a constant parameter in the governing equations.

Fuel consumption rates are a function of engine power for propellers, and engine thrust for jets.

propeller: $C = C_{S_p} P_E = C_{S_p} \frac{1}{\eta_p} F V = C_{S_p} \frac{1}{\eta_p} D V = \frac{C_{S_p}}{\eta_p} \frac{1}{2}\rho S V^2 C_x V$ (power delivered by engine = $\frac{1}{\eta_p}$ power delivered by propeller; thrust from propeller; aero velocity)

jet: $C = C_{S_J} F = C_{S_J} D = C_{S_J} \frac{1}{2}\rho S V^2 C_x$

$E, R = f(D, V) = f(h, V, \alpha, \underbrace{\text{aircraft geometry}}_{\text{fixed}})$

pilot can act on h, V, and α, BUT aircraft has to fly: constraint $L = W$

L_D only 2 degrees of freedom.

The pilot can make some choices of what will be kept constant during cruise. Easiest case is to keep 2 parameters constant, and vary the third one. This leads to 3 different cruise control laws:

- "operational": keep h and V constant → α will DECREASE during flight
- "Breguet": keep h and α constant → V will DECREASE during flight
- "varying altitude": keep α and V constant → h will INCREASE during flight

These choices lead to a total of

2 engines types (P/J) × 2 performances (E/R) × 3 control laws = 12 formulas

Note: it can be shown that all range/endurance formulas have a similar structure:

$$\text{PERFORMANCE} = Q \times F_M(w_m)$$

Q: quality coefficient independent of control law

$F_M(w_m)$: mass function depends solely on amount of fuel onboard compared to total mass

Let's cover one of the actual Breguet cruise control laws in detail (the other ones can be obtained in very similar ways, just expressing the chosen varying parameter as a function of the others using the expressions derived from the force balance)

Endurance of an aircraft equipped with jet engines (Breguet control law) → h and α constant, V allowed to vary

$$E = -\int_{m_i}^{m_f} \frac{dm}{C}$$

$$* \quad C = C_{s_J} \underbrace{D}_{\text{drag force}} = C_{s_J} \frac{1}{2} \rho S V^2 C_x$$

$$E = -\int_{m_i}^{m_f} \frac{dm}{C_{s_J} \frac{1}{2} \rho S V^2 C_x}$$

constant

To move forward, we need to figure out what is constant inside the integral

Assumption: α and h (so ρ) are kept constant

⇒ V will vary over time, so we need to replace it as a function of constant values and mass, the integrating variable.

Lift = Weight : $L = \frac{1}{2} \rho V^2 S C_z = mg$

$$\rightarrow V^2 = \frac{mg}{\frac{1}{2} \rho S C_z}$$

$$E = -\int_{m_i}^{m_f} \frac{\frac{1}{2} \rho S C_z}{C_{s_J} \frac{1}{2} \rho S C_x mg} dm$$

$$E = -\int_{m_i}^{m_f} \underbrace{\frac{1}{C_{s_J} g} \frac{C_z}{C_x}}_{\text{constant (since } \alpha \text{ is constant)}} \frac{dm}{m} = -\frac{1}{C_{s_J} g} \frac{C_z}{C_x} \int_{m_i}^{m_f} \frac{dm}{m}$$

$$E = -\frac{1}{C_{sf}\, g}\frac{C_z}{C_x}\left[\ln m\right]_{m_i}^{m_f} = \frac{1}{C_{sf}\, g}\frac{C_z}{C_x}\ln\left(\frac{m_i}{m_f}\right)$$

Q F_n

$$\boxed{E = \frac{1}{C_{sf}\, g}\frac{C_z}{C_x}\ln\left(\frac{1}{1-w_m}\right)}$$

↳ expressed using w_m:

$$w_m = 1 - \frac{m_f}{m_i}$$

We are interested in the maximum possible value for E, that is, the maximume time in flight

↳ we chose 2 parameters to keep constant: α and h

↑ pilot can choose α such that E will be maximum. the only part that depends on α is the ratio of force coefficients

⇒ find α such that C_z/C_x is maximum

↑ not involved here

Function to maximize : $\boxed{\frac{C_z}{C_x}}$

↳ C_z is the variable with respect to which C_z/C_z must be max.

Note: maximizing $\frac{C_z}{C_x}$ is equivalent to minimizing $\left(\frac{C_z}{C_x}\right)^{-1} = \frac{C_x}{C_z}$

Drag polar : $C_x = C_{x_0} + \frac{1}{\pi AR} C_z^{\,2}$

Extremum of $\frac{C_x}{C_z}$ is obtained when $\boxed{\frac{d}{dC_z}\left(\frac{C_x}{C_z}\right) = 0}$

$$\Rightarrow \frac{d}{dC_z}\left(\frac{C_{x_0} + \frac{C_z^{\,2}}{\pi AR}}{C_z}\right) = \frac{d}{dC_z}\left(\frac{C_{x_0}}{C_z} + \frac{C_z}{\pi AR}\right) = 0$$

$$\Rightarrow -C_{x_0}\frac{1}{C_z^{\,2}} + \frac{1}{\pi AR} = 0$$

$$\Rightarrow C_z\Big|_{f_{max}} = \sqrt{C_{x_0}\pi AR}$$

$$\Rightarrow f_{max} = \frac{1}{2}\sqrt{\frac{\pi AR}{C_{x_0}}}$$

Note: All Breguet formulas involve some function of C_z and C_x to maximize; but the exact function depends on type of engine:

$$\frac{C_z}{C_x}\ ,\ \frac{C_z^{\,3/2}}{C_x}\ ,\ \frac{C_z^{\,1/2}}{C_x}$$

The following table provides the expressions for all possible range/endurance formulas.

- Some formulas directly include the optimal expressions for C_z (*e.g.* Q_{EP_1}). However, others (say, Q_{EP_2}) only include the expression of C_z and C_x that needs to be maximized before the actual endurance and range (*i.e.* their max values) can be calculated. In those cases, C_z should be chosen so that the function of C_x and C_z involved in the formula is maximized.

Propeller

Perf	M	Quality factor Q	Mass Factor F_M
E	1	$Q_{EP_1} = \frac{\sqrt{3}}{4}\frac{\eta_P\sqrt{\rho}}{C_{s_P}\sqrt{2}g^{3/2}}(\pi \mathbf{AR})^{3/4}\sqrt{\frac{S}{m_i}}\left(\frac{3}{C_{x_0}}\right)^{1/4}$	$F_{M_{EP_1}} = \frac{\frac{1}{\sqrt{3}}+\sqrt{3}}{(1-w_m)^{1/4}}\arctan\frac{w_m}{\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right)\sqrt{1-w_m}}$
	2	$Q_{EP_2} = \frac{\eta_P}{C_{s_P}}\sqrt{\frac{\rho}{2}}\frac{1}{g^{3/2}}\sqrt{\frac{S}{m_i}}\frac{C_z^{3/2}}{C_x}$	$F_{M_{EP_2}} = 2\left(\frac{1}{\sqrt{1-w_m}}-1\right)$
	3	$Q_{EP_3} = \frac{\eta_P}{C_{s_P}g^{3/2}}\sqrt{\frac{S}{m_i}}\sqrt{\frac{\rho_i}{2}}\frac{C_z^{3/2}}{C_x}$	$F_{M_{EP_3}} = \ln\left(\frac{1}{1-w_m}\right)$
R	1	$Q_{RP_1} = \frac{\eta_P}{2C_{s_P}g\sqrt{kC_{x_0}}}$	$F_{M_{RP_1}} = 2\arctan\left(\frac{w_m}{2\sqrt{1-w_m}}\right)$
	2	$Q_{RP_2} = \frac{\eta_P}{gC_{s_P}}\frac{C_z}{C_x}$	$F_{M_{RP_2}} = \ln\left(\frac{1}{1-w_m}\right)$
	3	$Q_{RP_3} = \frac{\eta_P}{gC_{s_P}}\frac{C_z}{C_x}$	$F_{M_{RP_3}} = \ln\left(\frac{1}{1-w_m}\right)$

operational — Breguet — varying h

Jet engines

Perf	M	Quality factor Q	Mass Factor F_M
E	1	$Q_{EJ_1} = \frac{1}{2gC_{s_J}}\sqrt{\frac{\pi \mathbf{AR}}{C_{x_0}}}$	$F_{M_{EJ_1}} = 2\arctan\left(\frac{w_m}{2\sqrt{1-w_m}}\right)$
	2	$Q_{EJ_2} = \frac{1}{2gC_{s_J}}\sqrt{\frac{\pi \mathbf{AR}}{C_{x_0}}}$	$F_{M_{EJ_2}} = \ln\left(\frac{1}{1-w_m}\right)$
	3	$Q_{EJ_3} = \frac{1}{2gC_{s_J}}\sqrt{\frac{\pi \mathbf{AR}}{C_{x_0}}}$	$F_{M_{EJ_3}} = \ln\left(\frac{1}{1-w_m}\right)$
R	1	$Q_{RJ_1} = \frac{\sqrt{3}}{4}\frac{\sqrt{2}}{C_{s_J}\sqrt{\rho g}}(3\pi \mathbf{AR})^{\frac{1}{4}}\sqrt{\frac{m_i}{S}}\frac{1}{C_{x_0}^{3/4}}$	$F_{M_{RJ_1}} = \left(\frac{4}{\sqrt{3}}\right)(1-w_m)^{\frac{1}{4}}\arctan\frac{w_m}{\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right)\sqrt{1-w_m}}$
	2	$Q_{RJ_2} = \frac{1}{C_{s_J}\sqrt{g}}\sqrt{\frac{2}{\rho}}\sqrt{\frac{m_i}{S}}\frac{C_z^{1/2}}{C_x}$	$F_{M_{RJ_2}} = 2\left(1-\sqrt{1-w_m}\right)$
	3	$Q_{RJ_3} = \frac{1}{C_{s_J}\sqrt{g}}\sqrt{\frac{2}{\rho_i}}\sqrt{\frac{m_i}{S}}\frac{C_z^{1/2}}{C_x}$	$F_{M_{RJ_3}} = \ln\left(\frac{1}{1-w_m}\right)$

Q_{RJ_1}, Q_{RJ_2}, Q_{RJ_3} are expressed slightly differently here, but they are equal:

$\frac{C_z^{1/2}}{C_x}$ has just been replaced by its maximum value in Q_{RJ_1} (see next page)

Optimals of functions of C_z and C_x

Function of C_z and C_x to maximize	Optimal C_z	Value of function	Used in
$\frac{C_z^{3/2}}{C_x}$	$\sqrt{3 C_{x_0} \pi AR}$	$\frac{(3\pi AR)^{3/4}}{4 C_{x_0}^{1/4}}$	Endurance (PROP.)
$\frac{C_z}{C_x}$	$\sqrt{C_{x_0} \pi AR}$	$\frac{1}{2}\sqrt{\frac{\pi AR}{C_{x_0}}}$	Endurance (JET) Range (PROP.)
$\frac{C_z^{1/2}}{C_x}$	$\sqrt{\frac{1}{3} C_{x_0} \pi AR}$	$\frac{3}{4}^{3/4} \frac{(\pi AR)^{1/4}}{C_{x_0}^{3/4}}$	Range (JET)

5 Absolute and Sustentation ceilings

Both ceiling performance metrics are explored in detail in the first case study. Therefore, only a brief definition is provided here.

- **Absolute ceiling** The absolute ceiling is the highest altitude at which an aircraft can fly at a given, constant velocity given its engines. At the absolute ceiling, all power/thrust available is used to maintain speed and altitude.

  - typically evaluated at either the lift coefficient C_z optimal for cruise conditions OR the maximal realistic C_z for the aircraft.
    - note that if C_z is fixed, the velocity at which the aircraft will fly at the absolute ceiling is fixed too due to the sustentation equation
  - the absolute ceiling is linked to the ability of the engine to maintain Thrust = Drag.

- **The sustentation ceiling** is the highest altitude at which an aircraft can fly, regardless of the power/thrust its engines can provide.

  - determined for a given cruise velocity, or Mach #
  - corresponds to the situation where level flight would result in stall, no matter what the engines can provide
    - link to ability of surrounding air to apply sufficient lift force on aircraft.

Typically, $h|_{\text{sustentation ceiling}} > h|_{\text{absolute ceiling}}$